

Monetary Policy Theory

Chapter 23

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

This chapter applies the **AD–AS** framework to analyze how central banks conduct monetary policy in response to economic shocks.

Central questions we will address:

1. How should monetary policy respond to **demand** vs. **supply** shocks?
2. What limits the effectiveness of monetary policy in the **long run**?
3. What are the ultimate **sources of inflation**?
4. What happens when interest rates hit **zero**?

Outline

Monetary Policy Responses to Economic Shocks

Policy Lags and Activist Policy

Long-Run Limits of Monetary Policy

Sources of Inflation

The Zero Lower Bound and Unconventional Policy

The Central Bank's Objective

The primary goal of monetary policy is to minimize deviations of inflation and output from their target levels.

The Inflation Gap

$$\text{Inflation Gap} = \pi^{actual} - \pi^{target}$$

A positive gap means inflation is **too high**; a negative gap means it is **too low**.

The Output Gap

$$\text{Output Gap} = Y^{actual} - Y^{potential}$$

A negative output gap indicates the economy is operating **below capacity** (recession).

Two Types of Shocks, Two Different Challenges

Demand Shocks

- ▶ Move inflation and output in the **same direction**
- ▶ Policy can stabilize **both** simultaneously
- ▶ Example: decline in consumer confidence

Supply Shocks

- ▶ Move inflation and output in **opposite directions**
- ▶ Policy faces a **trade-off**
- ▶ Example: oil price spike

Key Insight

The nature of the shock determines whether the central bank faces a **dilemma** or a **divine coincidence**.

Response to an Aggregate Demand Shock

Policy Options

When a negative demand shock hits (e.g., a financial crisis reduces spending):

- ▶ **Option 1: No policy response** — let the economy self-correct
- ▶ **Option 2: Stabilization policy** — ease monetary policy to support output

Response to an Aggregate Demand Shock

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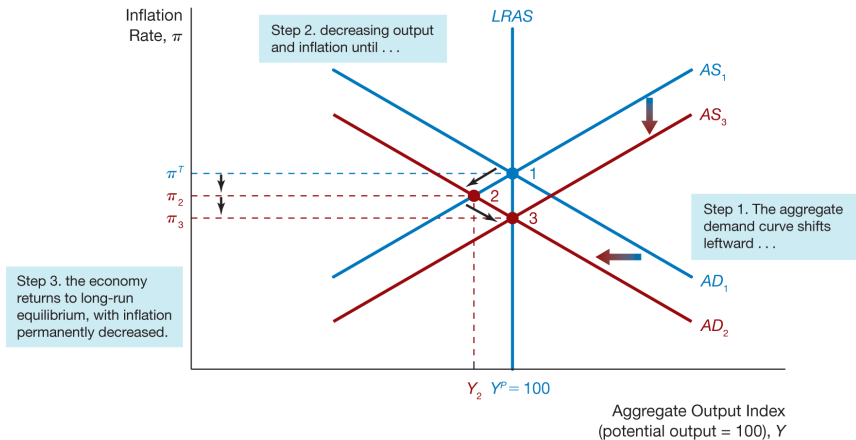
The Divine Coincidence (Blanchard & Galí, 2007)

With a **pure demand shock**, stabilizing inflation *automatically* stabilizes output.

There is **no trade-off** between price stability and output stability.

⇒ The central bank can “kill two birds with one stone.”

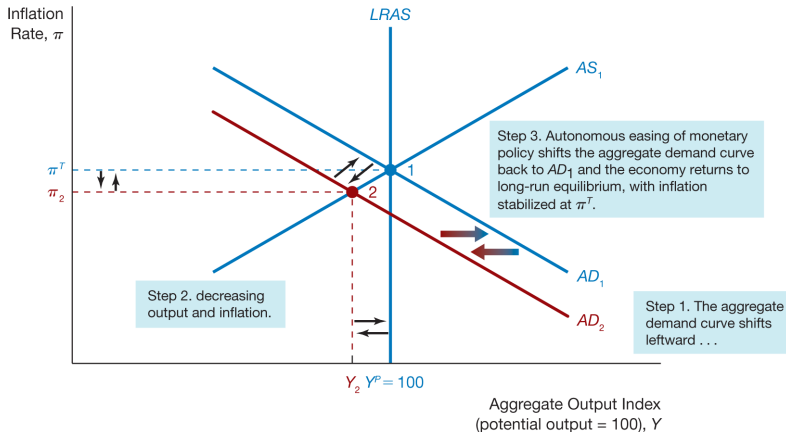
Aggregate Demand Shock: No Policy Response



Without intervention, the economy eventually self-corrects as AS shifts down, but the adjustment is slow and costly.

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Aggregate Demand Shock: Policy Stabilizes Output and Inflation



Expansionary monetary policy shifts AD back to AD_1 , restoring equilibrium at Y^P and π^T more quickly.

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Response to a Temporary Supply Shock

The Policy Dilemma

A temporary supply shock (e.g., an oil price spike) creates **stagflation**:

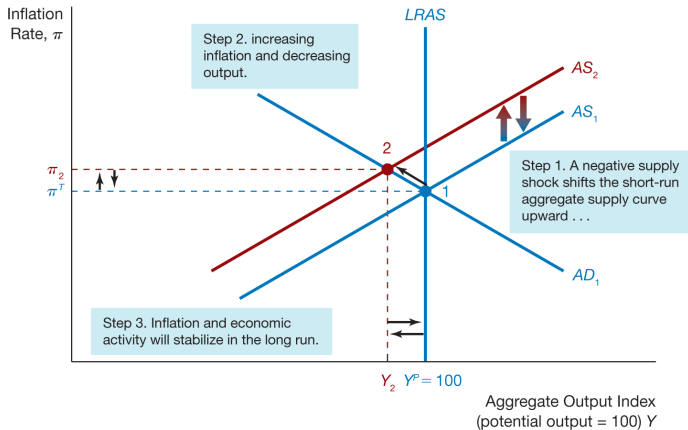
- ▶ Inflation **rises** (AS shifts up)
- ▶ Output **falls** (below potential)

The central bank **cannot** fix both problems at once.

Three Possible Responses

1. **No policy response** — wait for the shock to reverse
2. **Stabilize inflation** — tighten policy $\Rightarrow$ output falls further
3. **Stabilize output** — ease policy $\Rightarrow$ inflation rises further

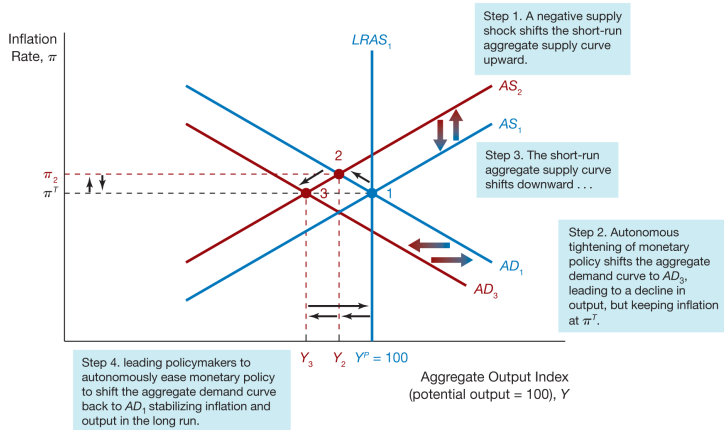
Supply Shock: No Policy Response



The economy moves to point 2 with higher inflation and lower output. Over time, AS shifts back and the economy returns to point 1.

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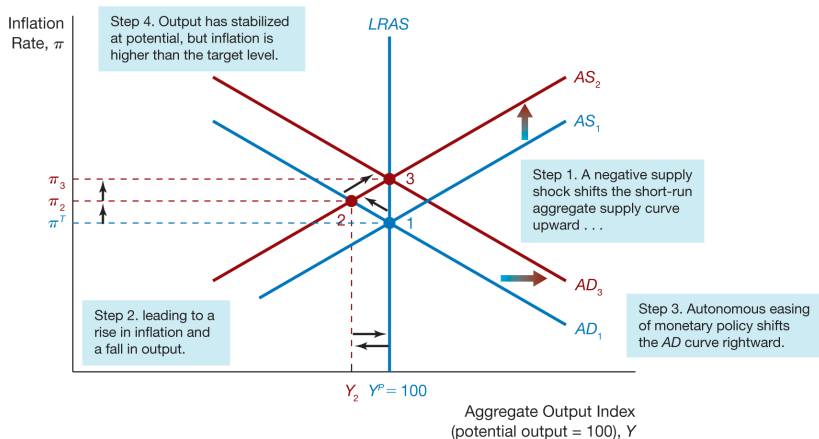
Supply Shock: Short-Run Inflation Stabilization



Tightening policy shifts AD left to AD_3 , keeping inflation at π^T but causing a **deeper recession** ($Y_3 < Y_2$).

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Supply Shock: Short-Run Output Stabilization



Easing policy shifts AD right to AD_3 , restoring output to Y^P but at the cost of **higher inflation** ($\pi_3 > \pi_2$).

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The Bottom Line: When Does the Trade-Off Arise?

Demand shocks dominate

Stabilizing inflation = stabilizing output

No trade-off (Divine Coincidence)

Supply shocks dominate

Stabilizing inflation $\neq$ stabilizing output

Short-run trade-off exists

Practical Implication

Central banks with a **dual mandate** (e.g., the Fed) must decide how much weight to place on each objective when supply shocks hit.

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The Activist vs. Nonactivist Debate

Activists

- ▶ Self-correction is **too slow**
- ▶ Prolonged recessions impose large **social costs**
- ▶ Wages and prices are **sticky downward**
- ▶ Government should **actively intervene**

Associated with: Keynesian tradition

Nonactivists

- ▶ Policy lags are **long and variable**
- ▶ Intervention may **destabilize** the economy
- ▶ The economy is **self-correcting**
- ▶ Government should follow **rules, not discretion**

Associated with: Monetarist / New Classical tradition

Policy Lags: Why Timing Is Difficult

Information Lags (Recognizing the Problem)

Data lag Time to collect and publish economic statistics (e.g., GDP is reported with a 1–3 month delay)

Recognition lag Time to confirm that a downturn is actually occurring (not just noise in the data)

Policy Lags: Why Timing Is Difficult

Information Lags (Recognizing the Problem)

Data lag Time to collect and publish economic statistics (e.g., GDP is reported with a 1–3 month delay)

Recognition lag Time to confirm that a downturn is actually occurring (not just noise in the data)

Action Lags (Implementing the Response)

Legislative lag Time for Congress to pass fiscal legislation (less relevant for monetary policy, since the Fed can act quickly)

Implementation lag Time to put the policy decision into effect

Effectiveness lag Time before the policy actually affects output and inflation (estimated at **6–24 months** for monetary policy)

Why Lags Matter: The Risk of Destabilization

The Core Problem

By the time a policy takes effect, the economy may have already moved to a **different phase** of the business cycle.



Implication for Monetary Policy

The Fed has a **shorter implementation lag** than fiscal policy (no legislative process), which is one reason monetary policy is the **primary tool** for stabilization.

Rules vs. Discretion

The Case for Rules

- ▶ Reduce **time-inconsistency** problems (the temptation to over-stimulate)
- ▶ Provide **transparency** and **predictability** to markets
- ▶ Example: **Taylor Rule**

$$i = \pi + r^* + a_{\pi}(\pi - \pi^T) + a_Y(Y - Y^P)$$

where r^* is the equilibrium real rate, $a_{\pi} > 0$, and $a_Y > 0$.

The Case for Discretion

- ▶ Rules cannot anticipate **every possible shock**
- ▶ The 2008 financial crisis required **unprecedented, creative responses**
- ▶ Rigid rules may **tie the central bank's hands** when flexibility is needed

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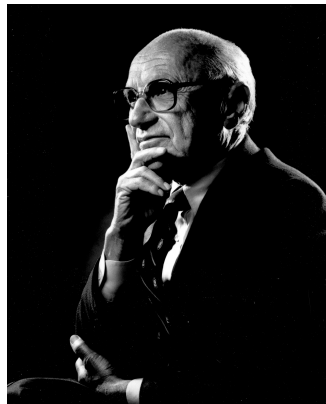
Inflation: Always and Everywhere a Monetary Phenomenon

Milton Friedman's Famous Dictum

"Inflation is always and everywhere a monetary phenomenon in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output." — Milton Friedman, 1963

In the AD–AS Framework

- ▶ In the **long run**, output returns to potential ($Y = Y^P$)
- ▶ The **only lasting effect** of monetary policy is on the **inflation rate**
- ▶ The central bank can choose any long-run inflation rate by positioning the AD curve



Milton Friedman (1912–2006), Nobel laureate 1976.

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Long-Run Neutrality of Money

Key Propositions

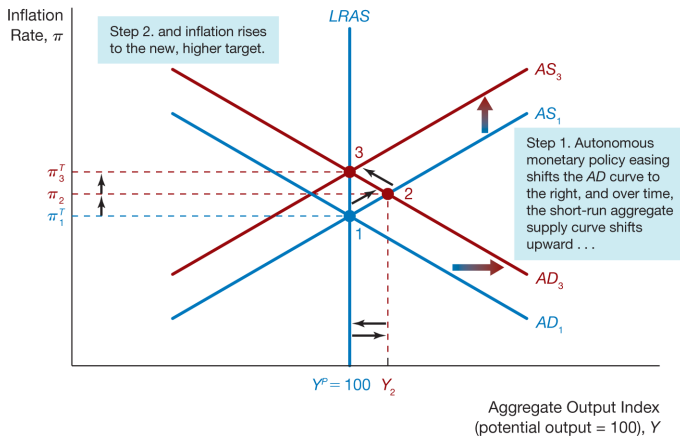
1. Monetary policy **cannot permanently** raise output above Y^P
2. Attempting to do so only produces **higher inflation**
3. In the long run, the AS curve is **vertical** at Y^P

The Temptation and Its Consequence

If the central bank tries to push output to $Y^T > Y^P$:

- ▶ It must **continuously** shift AD to the right
- ▶ This generates **ever-rising inflation**
- ▶ Expectations adjust upward, shifting AS up
- ▶ Output returns to Y^P , but inflation is permanently higher

A Rise in the Inflation Target



When the central bank raises its inflation target from π_1^T to π_3^T , the economy transitions through short-run adjustments before settling at the new, higher inflation rate.

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What Monetary Policy Can and Cannot Do

Objective	Short Run	Long Run
Stabilize output around Y^P	Yes	Yes
Raise output above Y^P permanently	Yes (temp.)	No
Choose the inflation rate	Yes	Yes
Lower unemployment below natural rate	Yes (temp.)	No

Bottom Line

Monetary policy is a powerful tool for **stabilization** and for determining the **long-run inflation rate**, but it cannot overcome the real constraints that determine potential output.

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Two Sources of Inflation

Cost-Push Inflation

Driven by **supply-side** factors:

- ▶ Negative supply shocks (e.g., oil price spikes)
- ▶ Wage increases exceeding productivity growth
- ▶ Rising raw material costs

Mechanism:

- ▶ AS shifts **upward** (left)
- ▶ Output falls, inflation rises
- ▶ If the central bank **accommodates**, a wage–price spiral can develop

Demand-Pull Inflation

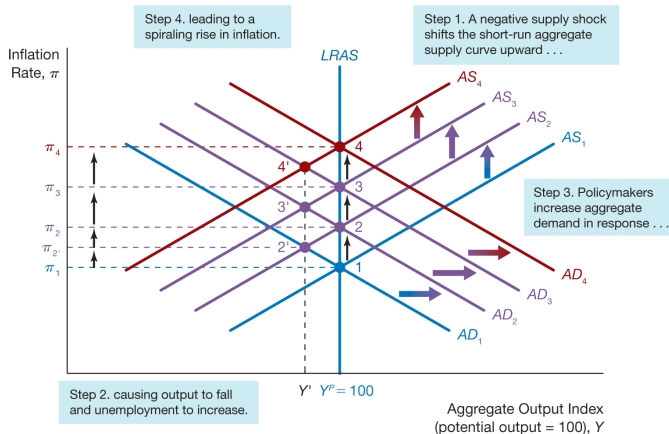
Driven by **demand-side** factors:

- ▶ Expansionary monetary policy
- ▶ Expansionary fiscal policy
- ▶ Surges in private spending

Mechanism:

- ▶ AD shifts **right**
- ▶ Output rises above Y^P
- ▶ Wages and prices adjust upward
- ▶ AS shifts up, output returns to Y^P at higher π

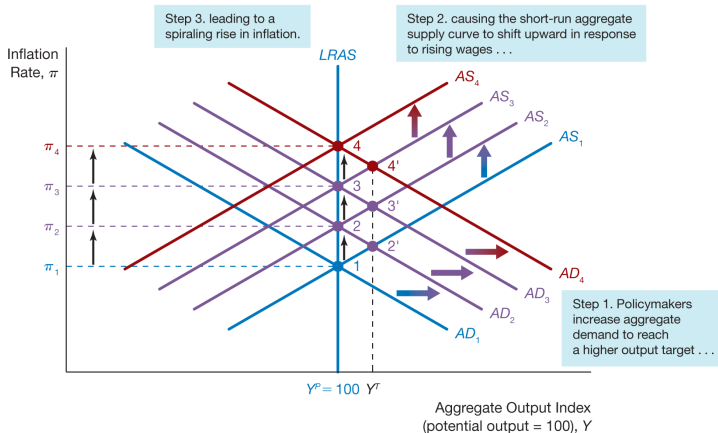
Cost-Push Inflation



Repeated supply shocks, accommodated by expansionary policy, produce a spiraling rise in inflation.

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Demand-Pull Inflation



Persistent expansionary policy pushes output above Y^P , triggering rising wages and a wage–price spiral.

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Can Budget Deficits Cause Inflation?

The Short Answer: Only If Monetized

Government budget deficits are inflationary **only** when the central bank **prints money** to finance them.

Bond-Financed Deficit

- ▶ Government borrows from the public
- ▶ No change in money supply
- ▶ **Not directly inflationary**
- ▶ May crowd out private investment

Money-Financed Deficit

- ▶ Central bank buys government bonds
- ▶ Money supply increases
- ▶ AD shifts right continuously
- ▶ **Inflationary**

Key: An **independent** central bank that refuses to monetize deficits is a crucial safeguard against fiscal-driven inflation.

Application: The Great Inflation (1965–1982)

Background

U.S. inflation rose from **below 2%** in the early 1960s to a peak of nearly **14% in 1980** — one of the most dramatic peacetime inflations in American history.

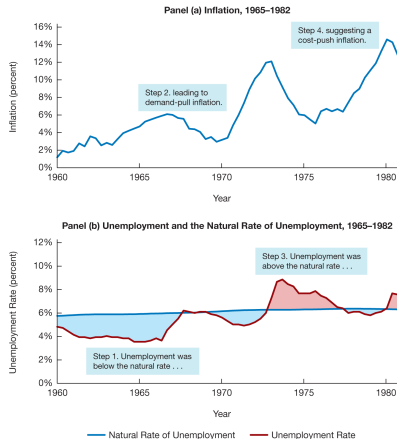
Phase 1: Demand-Pull (1965–73)

- ▶ Vietnam War & Great Society spending
- ▶ Unemployment fell **below** the natural rate
- ▶ The Fed **accommodated** fiscal expansion
- ▶ Inflation rose steadily

Phase 2: Cost-Push (1973–80)

- ▶ OPEC oil shocks (1973, 1979)
- ▶ Unemployment rose **above** the natural rate
- ▶ The Fed again accommodated
- ▶ Result: **stagflation**

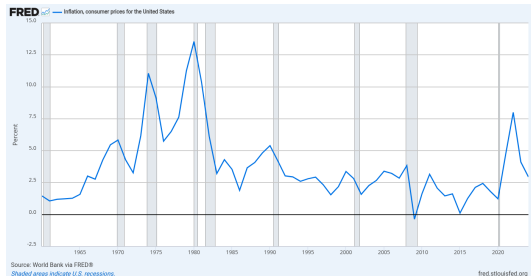
Inflation and Unemployment, 1965–1982



Inflation and unemployment in the United States, 1965–1982: demand-pull, then cost-push.

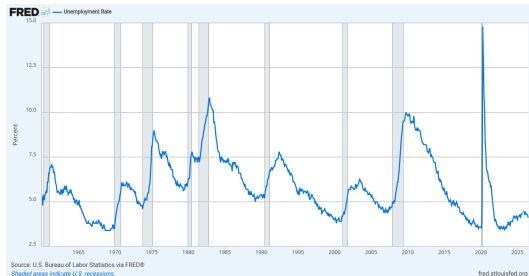
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The Great Inflation in the Data



U.S. CPI inflation, 1960–2024: the climb to nearly 14% in 1980 and the Volcker disinflation.

Source: FRED, Federal Reserve Bank of St. Louis



U.S. unemployment rate, 1960–2024: rising with each oil shock and peaking in 1982.

Source: FRED, Federal Reserve Bank of St. Louis

Phase 1 (1965–73): inflation rose while unemployment stayed **low** — demand-pull.
Phase 2 (1973–80): inflation **and** unemployment rose together — cost-push stagflation.

Lessons from the Great Inflation

What Went Wrong?

1. The Fed **underestimated the natural rate** of unemployment, believing it could sustain output above potential
2. The Fed **accommodated** supply shocks instead of anchoring inflation expectations
3. Political pressure made it difficult to **tighten policy** during recessions

Sustained inflation requires **monetary accommodation**: neither the fiscal expansion of the 1960s nor the oil shocks of the 1970s could have produced a decade of rising inflation had the Fed refused to accommodate them.

The Volcker Disinflation (1979–82)

Breaking the Spiral

Fed Chair Paul Volcker **raised interest rates sharply**, accepting a deep recession to break the inflationary spiral. By 1983, inflation had fallen to about **3%**.

⇒ This demonstrated that **credible commitment** to price stability is essential.



Fed Chair Paul Volcker (right) meets President Reagan in the Oval Office to discuss monetary policy, 1981.

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The Zero Lower Bound Problem

How Monetary Policy Normally Works

The Fed adjusts the **federal funds rate** (a nominal rate) to influence the **real interest rate**:

$$r = i - \pi^e$$

Lower $i \Rightarrow$ lower $r \Rightarrow$ more investment and consumption $\Rightarrow$ higher output.

¹Some central banks (ECB, BOJ, SNB) have experimented with slightly negative rates, hence the term “effective lower bound” (ELB).

The Zero Lower Bound Problem

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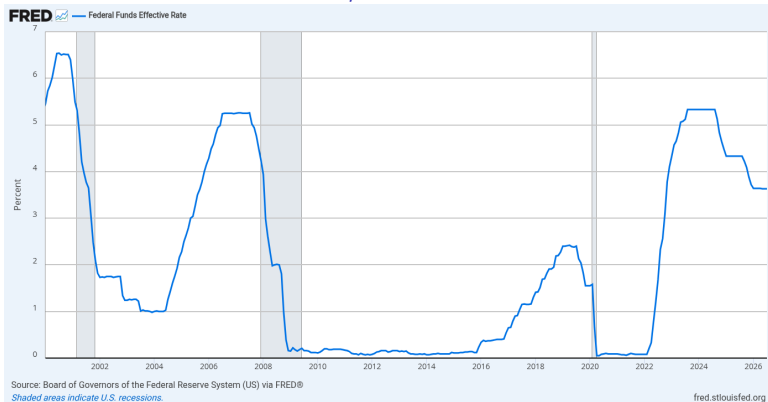
The Constraint

The federal funds rate is a **nominal** interest rate and **cannot fall below zero**.¹

When $i = 0$, the central bank has **run out of conventional ammunition**.

¹Some central banks (ECB, BOJ, SNB) have experimented with slightly negative rates, hence the term “effective lower bound” (ELB).

The Fed at the Zero Lower Bound, 2008–15 and 2020–22



Effective federal funds rate since 2000: stuck near zero for seven years after 2008 and again in 2020–21.

Source: FRED, Federal Reserve Bank of St. Louis

With $i = 0$, conventional policy has **run out of ammunition**; further stimulus must come from unconventional tools.

The AD Curve at the Zero Lower Bound

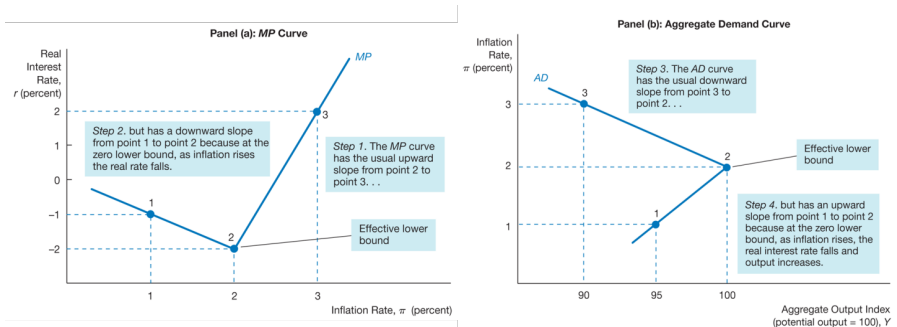
Normal Region (π is high enough)

- ▶ The MP curve is **upward sloping** (higher $\pi \Rightarrow$ higher r via Taylor rule)
- ▶ The AD curve is **downward sloping** (standard case)

At the ZLB (π is very low)

- ▶ The nominal rate is stuck at zero: $i = 0$
- ▶ If π^e falls, the real rate **rises**: $r = 0 - \pi^e$
- ▶ Higher r **reduces** spending and output
- ▶ The MP curve becomes **downward sloping**
- ▶ The AD curve develops an **upward-sloping kink**

Derivation of the AD Curve with an Effective Lower Bound



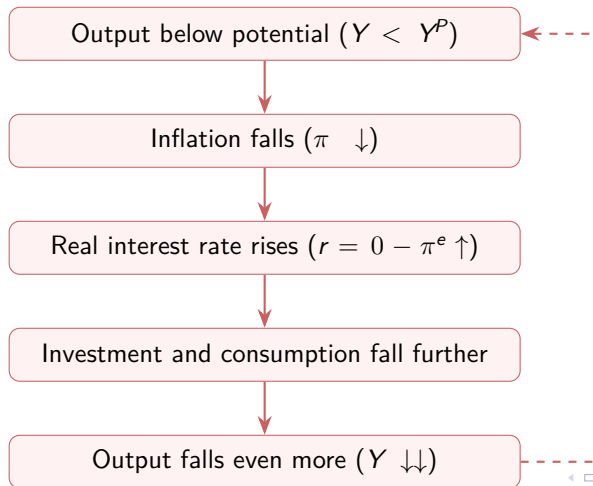
Panel (a): The MP curve has a kink at the ELB. Panel (b): The resulting AD curve is upward-sloping below the ELB threshold.

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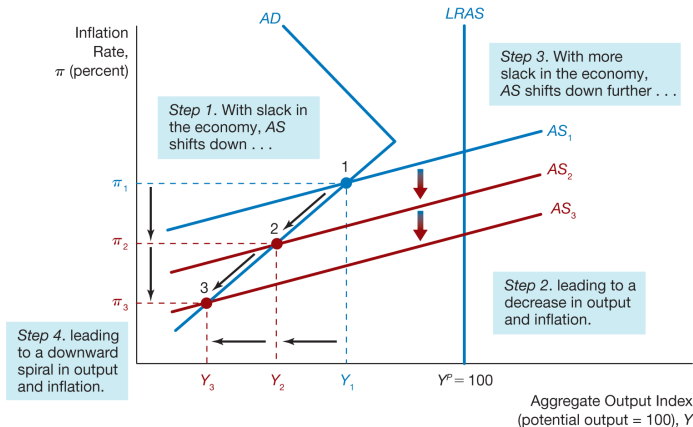
The Deflationary Spiral

Why the Self-Correcting Mechanism Breaks Down

At the ZLB, the normal self-correction process **works in reverse**:



The Absence of Self-Correction at the Effective Lower Bound



On the upward-sloping portion of AD, falling AS leads to **lower** output and **lower** inflation — a downward spiral.

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Unconventional Monetary Policy: The Toolkit

When conventional policy (lowering i) is exhausted, central banks turn to **unconventional tools**:

1. Liquidity Provision

- ▶ Expand lending facilities
- ▶ Reduce financial frictions
- ▶ Restore market functioning

2. Asset Purchases (QE)

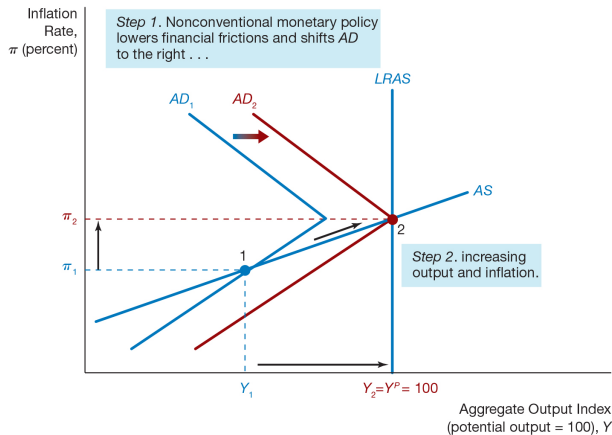
- ▶ Buy government & private bonds
- ▶ Lower long-term rates
- ▶ Reduce credit spreads

3. Forward Guidance

- ▶ Commit to low rates for extended period
- ▶ Raise inflation expectations
- ▶ Lower long-term rates

All three tools work by shifting the **AD curve to the right**, even when $i = 0$.

Response to Nonconventional Monetary Policy



Unconventional policy lowers financial frictions and shifts AD from AD_1 to AD_2 , increasing both output and inflation.

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How Forward Guidance Works

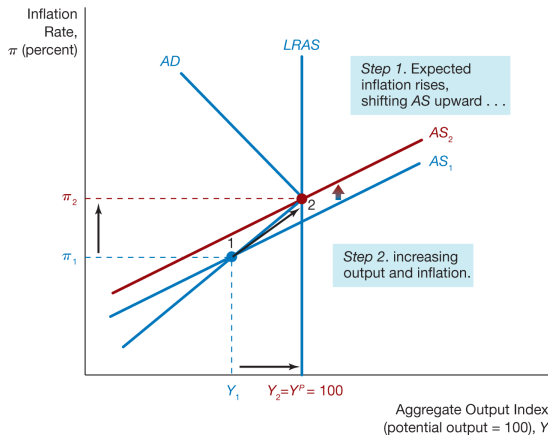
The Mechanism

The central bank announces: “We will keep interest rates at zero until [specific condition is met].”

Why This Helps

1. **Expectations channel:** If people expect rates to stay low, they expect **higher future inflation** $\Rightarrow \pi^e$ rises $\Rightarrow$ real rate falls
2. **Term structure channel:** Long-term rates are an average of expected future short-term rates; committing to low short rates **pulls down long rates**
3. **Both AD and AS shift:** Higher π^e shifts AD right (lower real rate) *and* shifts AS up

Response to a Rise in Inflation Expectations



Higher inflation expectations shift AS upward and (via lower real rates on the upward-sloping AD) increase output toward Y^P .

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Application: Abenomics and Japan (2013)

Background

Japan had been stuck in a **deflationary trap** since the 1990s. In 2013, Prime Minister **Shinzo Abe** launched an aggressive policy package.

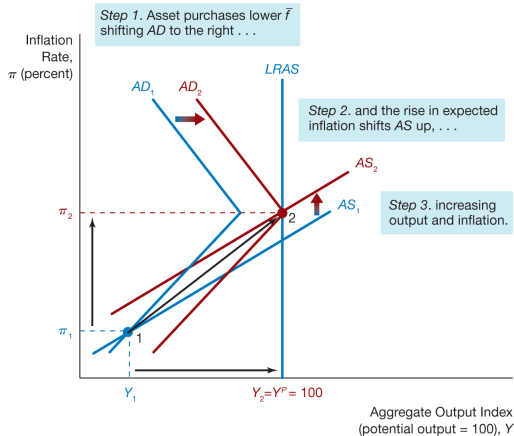
The “Three Arrows” of Abenomics

1. **Monetary easing:** Bank of Japan doubled its inflation target to 2% and launched massive QE
2. **Fiscal stimulus:** Increased government spending
3. **Structural reforms:** Deregulation and labor market reforms

Expected Macroeconomic Effects

- ▶ QE $\Rightarrow$ lower real interest rates $\Rightarrow$ AD shifts right
- ▶ Higher inflation target $\Rightarrow \pi^e$ rises $\Rightarrow$ AS shifts up
- ▶ Combined effect: output rises toward Y^P , escaping the deflationary trap

Response to the Shift in Japanese Monetary Policy in 2013



Asset purchases lower $\bar{r}$ and shift AD right; higher π^e shifts AS up. The economy moves from point 1 to point 2, with higher output and inflation.

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Chapter Summary

1. **Demand shocks** allow the central bank to stabilize inflation and output simultaneously (**Divine Coincidence**).
2. **Temporary supply shocks** force a **trade-off** between price stability and output stability.
3. **Policy lags** (data, recognition, implementation, effectiveness) complicate activist stabilization and can make policy **destabilizing**.
4. In the **long run**, monetary policy determines the **inflation rate** but cannot raise output above potential — money is **neutral**.
5. Inflation can be **cost-push** (supply shocks) or **demand-pull** (excess demand), but sustained inflation requires **monetary accommodation**.
6. At the **zero lower bound**, the self-correcting mechanism breaks down; central banks must use **unconventional tools** (QE, liquidity provision, forward guidance).

Key Equations to Remember

Inflation Gap

$$\text{Inflation Gap} = \pi^{\text{actual}} - \pi^{\text{target}}$$

Fisher Equation (Real Interest Rate)

$$r = i - \pi^e$$

At the ZLB: $i = 0 \Rightarrow r = -\pi^e$. If π^e falls, r **rises**.

Taylor Rule

$$i = \pi + r^* + a_{\pi}(\pi - \pi^T) + a_Y(Y - Y^P)$$

Provides a systematic framework for setting the policy rate.